

j4_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)

LEFT rail				RIGHT rail			
				+--[USB-C]--+			
I2C SDA (ADS x4 + keypad)	--	21				5V	
I2C SCL	--	22				GND	ground
DISP-L TX -> XIAO D7	--	17		TTGO		27	DISP-L RX <- XIAO D6
(free)	--	2		T-Display		26	DISP-R RX <- XIAO D6
AUX toggle	--	15		v1.1		25	DISP-R TX (face msgs)
EYE POP toggle	--	13				33	VENT toggle
(avoid: boot/flash)	--	12		[ST7789		32	LASER toggle
ground	--	GND		TFT on		39	(input only)
3.3V	--	3V3		back]		38	(input only)
5V	--	5V				37	(input only)
ground	--	GND				36	(input only)
				+-----+			

on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO0,
 GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23
 I2C bus: ADS_01 @ 0x48, ADS_02 @ 0x49, ADS_03 @ 0x4A, ADS_04 @ 0x4B,
 PCF8574 keypad_left @ 0x21, keypad_right @ 0x20
 toggles: INPUT_PULLUP, switch closes to GND (ON = LOW)

GPIO	Function
21	I2C SDA (ADS1115 x4, PCF8574 keypads x2)
22	I2C SCL
17	Serial1 TX to j4_display_left (XIAO D7)
27	Serial1 RX from j4_display_left (XIAO D6)
25	Serial2 TX to j4_display_right (XIAO D7, face-preset messages)
26	Serial2 RX from j4_display_right (XIAO D6, pot feed)
32	LASER toggle (INPUT_PULLUP, switch closes to GND)
33	VENT toggle (INPUT_PULLUP, switch closes to GND)
13	EYE POP toggle (INPUT_PULLUP, switch closes to GND)
15	AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)
34	Battery voltage sense (ADC input, read-only)
35	Screen-cycle button (TTGO built-in; data / MAC / connection status)

j4_controller -- ADS1115 pin diagrams

pot wiper to the A-pin; outer legs to 3.3V and GND

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ADS_01 @ 0x48 (ADDR -> GND)                four face pots, left bank
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | GND = address 0x48
| ALRT | not connected
| A0 | Eyebrow L pot wiper -> PCA9685 ch 6
| A1 | Eyebrow R pot wiper -> PCA9685 ch 7
| A2 | Basket Eyebrow L pot wiper -> PCA9685 ch 8
| A3 | Basket Eyebrow R pot wiper -> PCA9685 ch 9
+-----+

ADS_02 @ 0x49 (ADDR -> VDD)                both joysticks
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | VDD = address 0x49
| ALRT | not connected
| A0 | eyes joystick X wiper -> eyes pan servo (PCA9685 ch 3)
| A1 | eyes joystick Y wiper -> eyes tilt servo (PCA9685 ch 4)
| A2 | neck joystick X wiper -> neck L/R differential mix
| A3 | neck joystick Y (jaw) wiper -> neck L/R base height
+-----+

ADS_03 @ 0x4A (ADDR -> SDA)                four face pots, right bank
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | SDA = address 0x4A
| ALRT | not connected
| A0 | Nose pot wiper (up/down) -> PCA9685 ch 10
| A1 | Nose Basket pot wiper -> PCA9685 ch 11
| A2 | Bottom Eyelid L pot wiper -> PCA9685 ch 12
| A3 | Bottom Eyelid R pot wiper -> PCA9685 ch 13
+-----+

ADS_04 @ 0x4B (ADDR -> SCL)                neck-pivot + linear faders
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | SCL = address 0x4B
| ALRT | not connected
| A0 | neck-pivot pot wiper -> nP on the stepper link (0-3200)
| A1 | fader_left wiper -> Nose Basket (doubles ADS_03 A1)
| A2 | fader_right wiper -> iris (doubles j4_display_right IRIS)
| A3 | spare
+-----+
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